

The chalk hyporheic zone: a true ecotone?

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Abstract This review summarises the main ecotonal properties of chalk hyporheic zone (CHZ), using a holistic approach on both structure and functionality of this habitat. The corroborated results suggest that the CHZ represents a typically shallow habitat (approx. 40–50 cm deep), with a homogenous distribution of both fauna and chemistry between heads and tails of riffles (putative down and, respectively, upwelling zones). Despite being groundwater fed, the CHZ is equally influenced by surface waters, chemically and biologically. However, despite its shallowness, the CHZ plays a very important role in nutrients (re)cycling and in the energy flux towards river ecosystems, fuelling benthic food webs with surface-derived macronutrients and subsurface chemosynthetic C. Although groundwater variation influences strongly the river flow during winter, the effects on interstitial fauna dynamic are limited,

suggesting the active role of CHZ played in the stochastic events of down/upward migration of fauna across habitats. Overall, the CHZ represents a confined, but critical habitat in the structure and functionality of chalk river ecosystems and fulfils most properties of a true ecotone.

Keywords Chalk hyporheic zone · Ecotone properties · Groundwater · Chalk streams

Introduction

Chalk is a minor geological deposit in many European countries, reaching notable dimensions mainly in the southern part of the UK, and some areas of northern France and Belgium (Berrie, 1992; Howden et al., 2004). Chalk streams represent a special type of rivers: they are groundwater (GW) fed, with a biannual fluctuation of water level due the influence of subterranean unconfined phreatic layers, constant temperature and chemistry, low ratio depth/width, scarce gravel bars, low riffle-pool-riffle continuity and the tendency of depositing fine sediments, unless they are washed out by rare and uncommon local spates (Wood & Armitage, 1997; Sear et al., 1999). These rivers were considered as being “perched groundwater aquifers” (Shand et al., 1995), with their general chemistry and hydrology influenced by GW, but differing significantly from it (Sear et al., 1999).

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Beginning with the middle of last century, several hydrological, hydrochemical and ecological aspects of chalk streams were investigated mainly in the south of the UK, but most of these studies were focused exclusively on benthic habitats or surface water (SW) (Casey, 1981; Berrie, 1992; Wright, 1992; Wright & Symes, 1999; Harrison & Harris, 2002; Wright et al., 2004; Riley et al., 2009), despite the acknowledgement of these ecosystems as being GW-fed. More recently, the boundary between underlying aquifers and SWs has begun to blur, as the scientific community recognised the crucial importance of the chalk hyporheic zone (CHZ) in the wellbeing of these streams (Flynn et al., 2002; Griffiths et al., 2006; Ireson et al., 2006; Maurice et al., 2006; Pretty et al., 2006; Wheater et al., 2007; Wriedt et al., 2007; Lapworth et al., 2009; Allen et al., 2010).

The physico-chemical and ecological properties of the hyporheic zone as an ecotone summarised, recently by Williams et al. (2010), the following peculiarities:

- (1) The hyporheic zone as an ecotone represents a transition between two adjacent ecosystems.
- (2) The hyporheic zone has an extensive linear dimension and sometimes is narrower than the adjoining communities and habitats themselves (e.g. rivers), depending on the size and order of hydrological network.
- (3) The community inhabiting the hyporheic zone (hyporheos, *sensu* Boulton, 2000) contains many of the organisms belonging to the adjacent areas, but supports organisms characteristic of and often confined to this ecotone.
- (4) Species richness and densities are greater in the hyporheic zone than in adjacent areas due to an edge effect, related to enhanced diversity of resources and availability.
- (5) Physical and chemical parameters in the hyporheic zone show discontinuities in their concentrations between adjacent habitats.
- (6) Spatial extent of the hyporheic zone varies with the variability of SW and GW.
- (7) The species composition and physical and chemical parameters differ in downwelling and upwelling areas, because of heterogeneity of hydrologic exchange between SW/GW.
- (8) The main macronutrients (C, N and P) are being (re)cycled in the hyporheic zone, supporting a unique local food web, different from those of the bordering habitats.

Although the importance of CHZ in the interaction between SW and GW was stressed several decades ago (Hynes, 1983), it can be concluded that a holistic approach of the structure and functionality of this ecosystem is lacking currently. The attempt made in this review is to offer a more coherent and unifying framework of these ecosystems and to summarise their ecotonal properties as described above.

Hydrology and hydrochemistry of CHZ

The role of GW in supporting the relatively constant base flow during summer in chalk streams and the high flows during winter was the focus of much research (Grapes et al., 2005). Among the three main types of aquifers in the UK (chalk, limestone and Permo-Triassic sandstone), the chalk streams' flow is characterised by the narrowest Base Flow Index (BFI), hence dominated by GW flows (Sear et al., 1999) (Table 1). The SW–CHZ–GW mixture takes place at different scales. Sometimes, over small distances, chalk streams show a complex of gains and losses of surface waters along their reaches (Grapes et al., 2005; Griffiths et al., 2006), whereas over larger scales (i.e. catchment), the heads of GW system may be more important, as the hydraulic conductivity within the alluvial system overlaying the chalk aquifer is fuelling the SW flow (Lapworth et al., 2009) (Table 1). Besides local GW level variation, these hydrologic exchanges seem to be equally controlled by local lithology and topography (Lapworth et al., 2009). Local sand and clay deposits can limit the hydrologic exchange because of their poor permeability, whereas gravel deposits can be much better connected with the GW (Pretty et al., 2006). The local scale permeability can play, therefore, a crucial role in the magnitude and direction of SW/GW hydrological interaction and is equally influenced by the catchment topography, mainly by the location and permeability of dry valleys that enhance the accretion rates at the confluence with the main stem of the river (Grapes et al., 2005). The CHZ and underneath aquifers are therefore in hydraulic continuity, allowing water movement between these hydrologic compartments (Allen et al., 2010). The interchange of water between

Table 1 Summary of the main characteristics of CHZ as opposed to the general trend observed in the literature for the hyporheic habitats of rivers flowing through different geologies

Features	CHZ	Classic studies
Hydrologic exchange at catchment scale	GW-fed streams with a major phreatic component throughout their hyporheic zone (Sear et al., 1999)	Mainly surface water-derived hydrologic exchange throughout the hyporheic zone, GW influence less important (Boulton et al., 1998)
Hydrologic exchange at riffle scale	Down/upwelling areas not clearly defined in the heads/tails of riffles; the interstitial water in the shallow sediments influenced by surface water but overall a strong groundwater component (Davy-Bowker et al., 2006; Pretty et al., 2006)	Down/upwelling areas defined in the heads/tails of riffles; at riffle scale the surface water component in hyporheic habitats is even stronger than at catchment scale (Williams & Hynes, 1974; Williams, 1993)
Hydrochemistry	Heads and tails of riffles virtually identical from a chemical point of view (Davy-Bowker et al., 2006; Pretty et al., 2006; Pacioglu et al., 2016)	Heads and tails of riffles significantly different with respect to their chemistry (Danielopol, 1976; Dole-Olivier et al., 1993, 1994)
Nutrients cycling	Decrease of concentration of main macronutrients (C,N and P) with depth; CHZ can act as both source and sink for nutrients depending on the degree of eutrophication of rivers (Pretty et al., 2006; Pacioglu et al., 2016)	Similar pattern (Mulholland et al. 1997; Strauss & Lamberti, 2000; Baker et al., 2000; Borges et al., 2006)
Chemosynthesis	Methane brought to both CHZ and SW from groundwater, offering an important alternative source of C for interstitial and benthic food webs (Sanders et al., 2007; Trimmer et al. 2009a, b)	Not very thoroughly studied, but current findings suggests that the GW influence and hence the upward input of nutrients in both benthic and interstitial habitat decreasing with the values of Base Flow Index (Trimmer et al., 2012)
Dissolved oxygen	Very well-oxygenated environment (Davy-Bowker et al., 2006; Pretty et al., 2006; Pacioglu et al., 2012, 2016)	Varies, but generally with a tendency for smaller concentrations of dissolved oxygen (Mulholland et al. 1997; Strauss & Lamberti, 2000; Baker et al., 2000)
Frequency of stygobites	Low, mainly because of the glacial legacy (Davy-Bowker et al., 2006; Pacioglu et al., 2016; Tod & Schmid-Araya, 2009)	High, but with similar low values in the northern hemisphere (Danielopol, 1976; Dole-Olivier et al., 1993, 1994; Pacioglu, 2010)
Frequency of epigeal fauna	High (Davy-Bowker et al., 2006; Pacioglu et al., 2016; Tod & Schmid-Araya, 2009)	Medium- High (Danielopol, 1976; Dole-Olivier et al., 1993, 1994; Stanford & Ward, 1993)
Longitudinal distribution of fauna	Homogenous distribution of fauna in the heads and tails of riffles (Davy-Bowker et al., 2006; Pacioglu et al., 2016)	Very heterogeneous distribution of fauna in the down and upwelling areas: the former with a higher frequency of epigeal fauna, whereas the latter dominated by stygobites (Danielopol, 1976; Dole-Olivier et al., 1993, 1994; Stanford & Ward, 1993; Boulton et al., 1997, 1998)
Vertical distribution	Shallow habitats (in average approx. 40 cm deep) (Davy-Bowker et al., 2006; Pacioglu et al., 2016; Pretty et al., 2006; Tod & Schmid-Araya, 2009)	Much deeper (up to 1 m deep) (Danielopol, 1976; Dole-Olivier et al., 1993, 1994; Stanford & Ward, 1993; Boulton et al., 1997, 1998)
GW influence on fauna migration	Apparently very small; potentially CHZ represents a resilient habitat to such groundwater fluctuations (Pacioglu, 2011)	Increasing density of invertebrates in the interstitial habitats during floods (Dole-Olivier et al., 1994)
Influence of drought	Downward migration of invertebrates within the CHZ during periods of severe summer droughts (Stubbington et al., 2015; Wood et al., 2010)	Downward migration of invertebrates within the hyporheic zone during periods of drought (Williams et al., 2010)
Contribution with individuals to benthic food webs	Low (Tod & Schmid-Araya, 2009)	High (Boulton et al., 1998)
Invertebrate abundance	Extremely low (Davy-Bowker et al., 2006; Pretty et al., 2006; Tod & Schmid-Araya, 2009; Pacioglu et al., 2016)	Significantly much higher (Danielopol, 1976; Dole-Olivier et al., 1993, 1994; Stanford & Ward, 1993; Boulton et al., 1997, 1998)

Table 1 continued

Features	CHZ	Classic studies
Influence of eutrophication	Exchange in fauna composition with the degree of eutrophication (Pacioglu et al., 2016)	Similar trend (Ward et al., 1992; Malard et al., 1996; Stein et al., 2010; Hartland et al., 2011; Iepure et al., 2013)
Effects of colmation on fauna	Increase in OM accumulation directly with fine sediment intrusion and of deposit feeders abundance (like Oligochaeta) (Pacioglu et al., 2012)	Strong effects on the structure and functionality of hyporheos (Nogaro et al., 2010)

CHZ/SW occurs at a range of scales, most active mixing occurring at the level of a few tens of cm of the river bed, but with some SW/GW interactions to a depth of several metres (Allen et al., 2010). Close to the river, where the CHZ overlies the chalk aquifer, the interstitial heads are smaller than GW, resulting in an upward movement of subterranean water towards the river, whereas in other parts it is noticed a reverse pattern, with SW heads higher than CHZ, strongly suggesting a recharge of alluvial deposits from SWs. These findings are therefore consistent with the hypothesis that CHZ provides a significant subsurface flow system underneath the flowing channel (Wheater et al., 2007) (Fig. 1a).

The clear localisation of down/upwelling zones of hydrologic exchange between SW/GW in chalk streams is not always confined into the heads/tails of riffles, contrary to common knowledge (Harvey & Bencala, 1993; Jones et al., 1995) (Fig. 1b). The well-known structure of riffles with downwelling head and upwelling tail flow paths is not commonly encountered in chalk streams (Davy-Bowker et al., 2006). The reasons are not very well known, but factors like the heterogeneity of subsurface flow, differences in substrate composition, the lack of typical riffle-pool-riffle sequences (Sear et al., 1999) and the general upwelling tendency of GW flow may explain this apparent homogeneity at riffle scale with respect to precise localisation of water exchange (Hill & Limbourn, 1998; Davy-Bowker et al., 2006; Pretty et al., 2006). Therefore, the pattern of hydrologic exchange across SW-CHZ-GW interface at riffle scale is characterised by the interplay between the downward pressure exerted by stream water in the first tens of centimetres within the CHZ, but opposed by the negative pressure posed by the general upward movement of shallow phreatic layer (Fig. 1a). Moreover, these hydrologic

exchanges in chalk streams take place without a clear individualisation of exchange areas like the head and tail of riffles (Fig. 1a). This particular type of water exchange characteristic to CHZ is therefore significantly different to the one described in the literature for other type of geologies (e.g. alluvial deposits, limestone and greensand), where surface water enters the hyporheic zone through downwelling areas (heads of riffles) and displays interstitial water for some distance before upwelling into the surface (tails of riffles) (but see Boulton et al., 1998) (Table 1; Fig. 1b).

The GW influence on the chalk streams is illustrated not only on the patterns of water exchange, but on the water chemistry as well (Jarvie et al., 2005). However, the intensity of these effects is reduced with increasing distance from source, hence to irrelevant patterns at larger scales between BFI values in chalk streams and different physical and chemical parameters like water pH, conductivity, temperature, sodium, calcium, chloride and sulphates (Sear et al., 1999). If the exchange between SW/GW is further blurred by the physical and chemical characteristics of river water, the local areas of exchange between river and CHZ (the downwelling and upwelling zones) show sometimes contradicting results. A survey of the heads and tails of riffles in chalk streams showed that temperature (Davy-Bowker et al., 2006) and chloride concentrations (Pacioglu, 2011) over a vertical profile fluctuated directly with the river flow, indicating an increasing SW-CHZ exchange during winter, but significantly diminished during summer, when the rivers are characterised mainly by a constant baseflow. Seasonally driven differences in the physico-chemical properties of both surface and interstitial waters do exist in chalk streams, because of varying GW level (Grapes et al., 2005).

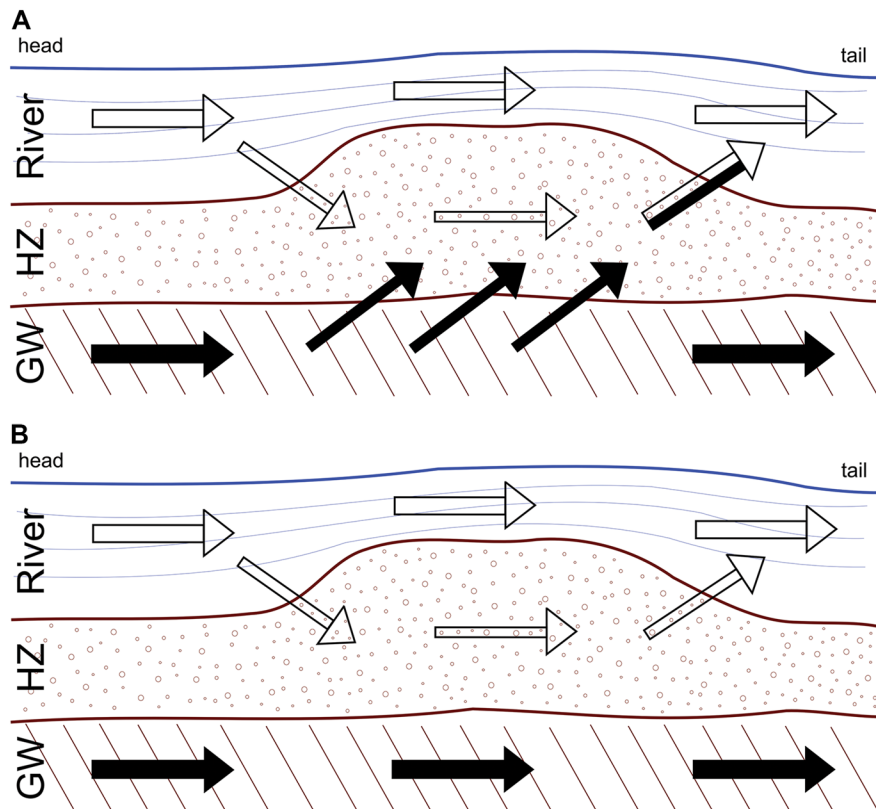


Fig. 1 **a** Pattern of water exchange at riffle scale in chalk streams. The hydrologic interplay is diffuse, with a general upwelling tendency of groundwater through interstices, but equally influenced in the upper sediment layers by stream water. Therefore, the CHZ at riffle scale consists of a mixture of both groundwater and stream water, without a clear individualisation of exchange areas. *White arrows* signify surface water, *black*

arrows groundwater and *double band arrows* a mixture of both ground and surface water reaching streams through CHZ. **b** Traditional view of water exchange at riffle scale between rivers and hyporheic zone. Surface water enters the hyporheic zone through downwelling areas (heads of riffles), displays interstitial water for some distance before upwelling into the surface (tails of riffles) (modified after Boulton et al., 1998)

Nutrients cycling within CHZ

The biogeochemical cycling of macronutrients (C, N and P) in CHZ is still a topic of hot debate and a deeper understanding of the behaviour of these nutrients is still missing (Jarvie et al., 2005; Pretty et al., 2006; Trimmer et al., 2009a, b).

The amount of organic matter decreases over depth in CHZ due to its attenuation by physico-chemical processes, adsorption onto clays (Lapworth et al., 2009) or because of its uptake by microbes (Trimmer et al., 2009a). Because the C sources can vary due to their chemical composition, the attenuation of labile C in chalk streams can occur in the first centimetres of sediment (Lapworth et al., 2009) (Table 1; Fig. 2). Clay surfaces, naturally occurring in CHZ, can act as both sink and source of labile C under changing

temporal conditions (Marmstrong & Barlocher 1989). Previous studies on methane production in the CHZ (Trimmer et al., 2009a, b) showed CO₂ efflux to be greater than the methane carbon equivalent due to the reduction of dissolved oxygen, suggesting that some oxidation processes are taking place via alternative electron acceptors (e.g. nitrate and sulphate). It was observed that nitrate concentration in CHZ decreases in the first 20 cm of sediment (Pretty et al., 2006; Pacioglu et al., 2016), which is in accordance with the previous studies undertaken on other eutrophic rivers (Mulholland et al., 1997; Baker et al., 2000; Strauss & Lamberti, 2000; Borges et al., 2006) (Table 1; Fig. 2). Pretty et al. (2006) noticed a patchy decrease of nitrate for the CHZ and attributed this to “pockets” of intense denitrification inside the sediments. Therefore, it is possible that the reduction of both C and N compounds

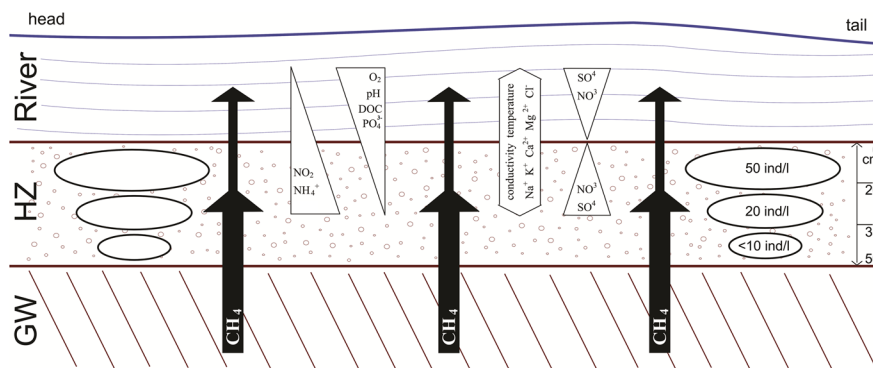


Fig. 2 Simplified schematic diagram of hydrochemistry, macronutrients cycling and fauna distribution with depth between heads and tails of chalk streams (riffle scale). Normally positioned *rectangular triangles* denote an increase with depth of nitrites and ammonia concentrations, whereas *inverted rectangular triangles* a decrease with depth of pH, dissolved oxygen, DOC and phosphate concentrations. *Double headed arrows* signify similar values for conductivity, temperature, sodium, potassium, magnesium, calcium and chloride concentrations at surface water–CHZ interface, whereas *opposed*

isosceles triangles the variability of nitrate and sulphate concentrations across the interface of these two habitats directly with the degree of eutrophication of chalk streams. *Black arrows* denote upwelling groundwater through interstices, bringing ancient methane that is subsequently transferred by bacteria into hyporheic and benthic food webs. The relative abundance of invertebrates (number of individuals/l) is decreasing with depth and shows a homogenous distribution between heads and tails of riffles

in CHZ, along with potential nitrification, methane oxidation and other biogeochemical reactions could result in a mosaic of “pockets”, where the oxic gradient changes rapidly towards anoxia over very small spatial scales (Baker et al., 2000; Maurice et al., 2006).

Consequently, the general high concentration of interstitial dissolved oxygen observed in the CHZ is unexpected (Pretty et al., 2006; Pacioglu et al., 2012, 2016), especially because chalk rivers are GW-fed, and intuitively, this element should be one of the first chemicals to be reduced in these habitats. Instead, although dissolved oxygen decreases on the vertical profile compared to SWs, high concentrations (4–6 mg l⁻¹) remain in the CHZ up to 35 cm depth (Pretty et al., 2006; Pacioglu et al., 2012, 2016) (Table 1; Fig. 2). Pretty et al. (2006) and Headworth et al. (1982) suggested that the CHZ is an environment highly connected to both SWs and GWs, powerfully squeezed between two well-oxygenated thoroughfares. The high connectivity between SW/GW could be responsible for the lack of a consistent gradient noticed between these two habitats with respect to conductivity, potassium, magnesium, calcium and sodium concentrations (Sear et al., 1999; Pacioglu et al., 2012, 2016) (Fig. 2). Phosphate levels in CHZ are generally low in CHZ (Lapworth et al., 2009; Pacioglu et al., 2016) and the greatest changes occur in

the first 50 cm of sediment (Lapworth et al., 2011). It was suggested that abiotic bonding in the sediment (i.e. coprecipitation with Ca ions) together with their potential biological uptake (Jarvie et al., 2005; Lapworth et al., 2011) are the possible pathways of phosphorous cycling in the CHZ. Because of active biogeochemical cycling in the upper sediment layers, CHZ can act as both source and sink of macronutrients for the river ecosystems (Jarvie et al., 2005; Pretty et al., 2006; Trimmer et al. 2009a, b; Pacioglu et al., 2016).

Structure and functionality of fauna in CHZ

The taxonomic composition of the hyporheos in the CHZ is very much of epigeal nature and stygofauna is very rare (Davy-Bowker et al., 2006; Tod & Schmid-Araya, 2009; Pacioglu et al., 2012, 2016). This invertebrate community contrasts strongly with the structure of hyporheos from other type of geologies, like alluvial deposits along big streams in Europe or big meandered rivers in the USA, where the stygofauna represents an important component (Danielopol, 1976; Dole-Olivier et al., 1993; Stanford & Ward, 1993) (Table 1). However, the stygofauna in CHZ is not significantly represented because generally speaking, the stygofauna from northern Europe (where

chalk streams mainly occur) was severely impoverished compared to southern and central Europe during the Pleistocene glaciations, that wiped out most GW dwelling fauna, excepting a handful of species that were old inhabitants of British Isles and that survived the latest glacial events (McInerney et al., 2014). Moreover, the low abundance of typical stygobitic macroinvertebrates in the CHZ and the high frequency of epigeal species is in accordance with the hydrological and hydrochemical findings (see above), suggesting that these interstitial habitats are ecologically strongly connected to SWs (Pretty et al., 2006). More thorough studies, including the permanent meiofauna (Tod & Schmid-Araya, 2009; Pacioglu et al., 2012, 2016), found only one species of stygobitic microcrustaceans (*Parastenocaris* gr. *fontinalis*) and few stygophilic microcrustaceans in the CHZ.

Although one of the characteristic of each ecotone is the presence of taxa representative for and confined to this habitat, the taxonomic investigations found almost no such cases for the interstitial habitats (Dole-Olivier et al., 1993). This is perhaps explained by the dynamic nature of this ecotone, which allows much less adaptation of specific taxa (Pacioglu, 2010). Some species of water mites have been described as being endemic and it is possible they represent the only group of invertebrates confined to and characteristic of this habitat (i.e. hyporheobionts, sensu Motas, 1962; Gledhill, 1973). However, taxonomic investigations of water mite communities in CHZ of four lowland streams discovered only typical epigeal taxa (Pacioglu et al., 2016).

The longitudinal distribution of chalk hyporheos in the heads and tails of riffles showed no significant differences between these two microhabitats (Davy-Bowker et al., 2006; Pacioglu, 2011) (Table 1). Previous studies suggested that in GW-fed rivers (including chalk), in contrast to rivers flowing through other geologies (Dole-Olivier et al., 1994), the downwelling zones are rare and the interstitial water flow is dominated by a general upwelling tendency, resulting therefore in a homogeneous interstitial habitat (including physico-chemical parameters) (Hill & Limbourn, 1998; Pretty et al., 2006; Abesser et al., 2008). Given the rarity of archetypal cross-sectional structure of riffles (with downwelling head and upwelling tail flow paths) in chalk streams (see above), the taxonomic composition of chalk

hyporheos at riffle scale is very homogenous (Table 1; Fig. 2). The vertical distribution of fauna is limited to the first 40–50 cm, with virtually no species present below this depth, excepting stygofauna (Davy-Bowker et al., 2006; Pacioglu, 2011) (Fig. 2). The abundance of hyporheos decreases with depth and shows no particular preference of both macroinvertebrates and meiofauna for certain depths in CHZ (Davy-Bowker et al., 2006; Pacioglu et al., 2016) (Table 1). This typical shallowness of CHZ could be again determined by the peculiar geomorphology and surface/subsurface water exchange of chalk streams, upwelling tendency of GW through interstices, leading to a habitat effectively squeezed between two well-extended dynamic habitats: the river and the unconfined phreatic layers (Pretty et al., 2006), therefore strongly reducing the potential for a vertical extension of this habitat (Fig. 2).

The fluctuation of GW and its influence on the SW flow during both summer and winter showed no clear patterns with respect to hyporheos abundance in four lowland chalk streams (Pacioglu, 2011), suggesting a mixture of taxa with different biological and reproductive traits that are very resilient to GW variation over a year. Recent studies showed that during summer droughts, some macroinvertebrates use the CHZ as a refuge and that not only the reduction of baseflow is an important driver of downward migration, but also the increased temperature and decreased dissolved oxygen concentrations associated with it (Wood et al., 2010; Stubbington et al., 2015). Moreover, the stygofauna composition from shallow subsurface habitats has been found to be significantly influenced by increasing GW water level during winter (Gledhill & Ladle, 1969; Gledhill, 1977). All these findings corroborated lead us to conclude that an apparent lack of GW influence on hyporheos dynamic (Pacioglu, 2011) is in fact determined by stochastic vertical migration events of invertebrates between two major thoroughfares bordering the CHZ: the rivers and shallow phreatic waters (Table 1; Fig. 2).

Another peculiarity of chalk hyporheos is represented by its extremely small abundance (in average 30 individuals l^{-1} , Pacioglu, 2011). The extremely low abundance of chalk hyporheos is potentially controlled by the confined vertical development of the habitat itself (approx. 40–50 cm) and the large quantity of fine sediments characterising the sediment composition of CHZ (Pretty et al., 2006) that is of

significant control on the size and availability of interstitial chalk habitat (Wood et al., 2010) and hence of the overall hyporheos abundance (Table 1; Fig. 2).

The structure of CHZ food webs, secondary production of the chalk hyporheos and its role in recycling the main macronutrients, as the influence of SW/GW on the transfer of energy flow between the bordering habitats into this habitat still awaits further investigations. The only study that we are aware of that tackled certain aspects of these issues suggested that the difference in size between macroinvertebrates and meiofauna is too great for the latter to contribute substantially to the overall secondary production in the epibenthic habitats and the first 10 cm of sediment from the CHZ (Tod & Schmid-Araya, 2009). Moreover, given limited vertical distribution of this habitat and the extremely low abundance of hyporheos (Fig. 2), it is possible to speculate that the role of CHZ in supporting benthic secondary production and epigeal food webs in terms of migrating taxa is limited, as suggested before with respect to shallow interstitial habitats (Findlay, 1995; Boulton et al., 1997) (Table 1). Still, the role of an ecotone in supporting the structure and functionality of food webs from adjacent ecosystems is not only through the migration of species, but also equally through the supply with nutrients for the energy flow (Williams et al., 2010). Despite being limited in space, the role of CHZ as both a sink and source for macronutrients was emphasised by several studies before (see above). The concentration of dissolved inorganic carbon in chalk streams is very high compared to other types of streams that are not GW-fed (Trimmer et al., 2009a). The $\delta^{13}\text{C}$ values of inorganic C in water, methane-oxidising bacteria, basal food resources, primary and secondary consumers in chalk streams are generally significantly smaller than similar food web components from streams that are not or only limited GW-fed (Trimmer et al., 2012). Therefore, it was suggested that besides autochthonous (photosynthetic) and allochthonous (detrital) organic matter source, the GW could subsidise with at least 30% of the required C of benthic consumers like Glossosomatidae and Goeridae caddisflies, dominant consumers in both benthic and CHZ (for the latter comprising early instars) habitats (Trimmer et al., 2009a, 2012). Given all these findings, one could speculate that in the CHZ, a completely heterotrophic-based ecosystem, methane and chemosynthetic produced C can provide a very

important energy source for the entire hyporheic food web (Table 1; Fig. 2). Since it is very likely that this ancient methane is brought by the underneath GW (Sanders et al., 2007), the important role of CHZ as an ecotone in subsidising the benthic food webs with nutrients is apparent.

Response of chalk hyporheos to eutrophication and colmation

Anthropogenic influences such as diffuse/point pollution with nitrate (Howden & Burt, 2009) or phosphorus (Casey & Newton, 1973; Hanrahan et al., 2001) in the case of rural catchments, and pesticides, heavy metals and households untreated waters (Flynn et al., 2002; Snook & Whitehead, 2004) for urban catchments are affecting both chalk streams and their adjacent CHZs in the UK (Jarvie et al., 2005; Pacioğlu et al., 2016). Besides eutrophication, the chalk streams are increasingly affected by colmation (Collins et al., 2013) and chalk rivers are particularly susceptible to fine sediment accumulation because they have stable non-episodic flow regimes which are increasingly impacted by water abstraction and drought (Collins & Walling, 2007). The deposition of fine sediments has profound effects on the local biogeochemistry of sediments (Nogaro et al., 2010) and on biota (Jones et al., 2015).

Recent studies focused on the relationship of chalk hyporheos with anthropogenic disturbances, mainly colmation and eutrophication (Jones et al., 2015; Pacioğlu et al., 2012, 2016). The findings suggested that the chalk hyporheos responds differently to these two disturbances. Communities characteristic to moderately eutrophic agricultural streams, but highly affected by colmation, are dominated by larval caddisflies and riffle beetles, whereas the hyporheos from urban chalk streams (highly eutrophic but with a moderate content of fine sediments) is instead dominated by species of copepods and ostracods, generalist taxa frequently encountered in the interstitial habitats affected by eutrophication and urbanisation (Ward et al., 1992; Malard et al., 1996; Stein et al., 2010; Hartland et al., 2011; Iepure et al., 2013) (Table 1). Field experiments, investigating solely the effect of colmation on chalk hyporheos, discovered that both at patch (head of riffles) and riffle scale, the fine sediment embedded in the CHZ led to an increase

in the content of organic matter and either increased the abundance of deposit feeding groups (i.e. Oligochaeta) or affected the vertical distribution of several taxonomic groups, although without a specific effect on a particular group (Pacioglu et al., 2012; Jones et al., 2015) (Table 1).

CHZ: does it fulfil the requirements of an ecotone?

Being an area not easily visualised or not easily sampled, the biotic and abiotic ecotonal characteristics of the hyporheic zone have been suggested but rarely demonstrated empirically (Pacioglu, 2010; Williams et al., 2010). The edge effect, well known for ecotonal areas, was suggested for hyporheic habitats (Stanford & Ward 1993; Williams, 1993), predicting a higher diversity of hyporheos than benthic habitats (Williams & Hynes, 1974). Nevertheless, the chalk hyporheos does not provide a higher diversity compared to benthic habitats, because its main components (meio- and macrofauna) are of epigeal origin and only occasionally supplemented by stygofauna characteristic to shallow GW (Table 1). Therefore, it can be considered that the CHZ does not necessarily provide a clear edge effect between the bordering habitats and the main reason is potentially because of its limited vertical development (approx 40–50 cm deep) and confined interstitial space, that in turn is restricting the abundance of the hyporheos and a certain homogeneity at riffle scale with respect to their spatial distribution (between the head and tail of riffles) (Table 1; Fig. 2).

Despite being a spatially confined habitat, its role in the biogeochemical cycle of macronutrients and subsidy of benthic food webs with nutrients (some of them derived from chemosynthesis) is nevertheless important. Therefore, the conclusion is that although limited and with an impoverished fauna, the main role of CHZ as an ecotone is mainly based on its role as an important subsidiary with nutrients (Table 1).

The hyporheic zone has been suggested as being a resilient ecotone and changing its dimensions with the variation of both GW and SW (Malard et al., 2002; Williams et al., 2010). Although direct measurements of the vertical extension dynamic of CHZ were never undertaken, certain fluctuations in its vertical and lateral development are to be expected as a consequence of biannual GW variation that sustains the

summer base flows and winter high flows, respectively. If to account for the unexpected summer droughts and downward migration of fauna into the interstitial *milieu* directly with the reduction of SW level, the dynamic role of the CHZ as an ecotone should be acknowledged (Table 1).

In conclusion, the CHZ represents a dynamic, albeit spatially limited transition zone between the rivers and subsurface ecosystems and that fulfils most of the requirements of an ecotone. The interstitial chalk habitats represent an extremely important ecosystem, crucial for the wellbeing of chalk streams and that mediates important biogeochemical processes at the SW/GW interface.

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